

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

When the Plant Network Drops: Comms Faults on the Floor

The line trips, the HMI (Human-Machine Interface, the operator screen) throws a red comms banner, and the controller shows a module fault. Everybody's first instinct is to swap a switch or re-pull a cable. Nine times out of ten that's the wrong first move, because the network already wrote down what happened and nobody read it. This is my field method for chasing comms dropouts across the mix of protocols you actually find on a floor — EtherNet/IP, Modbus TCP, PROFINET, PROFIBUS, DeviceNet — starting with the checks that cost nothing and keep you out of an energized panel until you have a reason to open one.

I'm writing this vendor-neutral. I run into these faults most on jobs where the network was built by one integrator, the drives by another, and the field devices got added over five years by three different hands. That handoff is where the ghosts live.

A quick note on what "the network" means here so we're speaking the same language. **EtherNet/IP** (Ethernet Industrial Protocol) is the CIP-over-Ethernet protocol used by Allen-Bradley/Rockwell and many others — it is *not* the same thing as generic office Ethernet, even though it rides the same copper and fiber. Where I use a term the first time I'll spell it out, because at 2am definitions matter more than jargon.

Read the evidence before you touch anything

The single habit that separates a 20-minute fix from a two-shift outage: **read the controller's comms/module fault bits and the managed switch's port statistics and timestamps before you swap any hardware.** The network keeps counters. Those counters are your black box recorder.

On the controller side, pull: - Module fault codes / connection status bits for each I/O and device connection. On a Rockwell ControlLogix/CompactLogix that's the module properties Connection tab and the controller fault log; on a Siemens S7 it's the diagnostic buffer (the timestamped fault log inside the CPU). Get the *timestamp* of the fault, not just that it faulted. - Connection timeout / connection loss counts. A connection that dropped and re-established leaves a count and usually a time.

On the managed switch side, pull the per-port statistics. You are hunting for: - **CRC** (Cyclic Redundancy Check — the error-detecting checksum on every Ethernet frame) errors and alignment errors. CRC errors climbing on one port point at a physical or noise problem on that specific run. - In/out errors, discards, and collisions (collisions on a modern full-duplex link mean a duplex mismatch — more on that below). - Link flap count and last-flap timestamp. A port that has flapped 400 times since Monday is telling you exactly where to look.

Line up the controller fault timestamp against the switch port flap timestamp against, if you have it, the drive event log. When three logs agree on a time, you've found your segment before you've touched a screwdriver. If the fault time matches the moment a specific VFD (Variable Frequency Drive) ramps, you've very likely found your cause too.

Write the counters down with the time you read them. Then, later, read them again. **A counter is only useful as a rate.** "500 CRC errors" means nothing. "500 CRC errors that became 900 in ten minutes while the mixer VFD ran" means everything.

| Safety before hands-on work

Everything above is read-only from an operator terminal or an engineering laptop on a mirror/SPAN port. You can do all of it without opening a panel. Do that work first.

When the trail leads you to hands-on work — reseating a connector inside a panel, ringing out a cable at the drive, checking a shield ground — the hazard changes.

- **Energized-panel caution.** Control panels carry 120/240/480 VAC and, on the drive side, higher. Arc flash is real. If you are opening an enclosure to probe, follow your site's electrical safety program: correct PPE, and where the task allows it, put the equipment in a verified zero-energy state and apply LOTO (Lockout/Tagout) before hands-on contact. Verify absence of voltage; don't trust the disconnect handle.
- **Stored energy on drives.** A VFD holds a charged DC bus after power-down. Do not open a drive and touch bus terminals on the assumption that "power is off." Wait out the manufacturer's stated discharge time — check the drive's own label/manual, commonly several minutes — and verify the bus is below the safe threshold with a meter before hands-on work.
- **Rotating and stored-energy equipment.** Any pump, mixer, conveyor, or fan on the far end of that I/O drop must be at a verified zero-energy/LOTO state before you work on it or near it. A comms fault that unexpectedly re-establishes can command a start.

Hazardous-area and gas caveat, only if it applies to where the cable actually runs: if any part of this network passes through a biogas/digester area, a pump vault, a sump, a silo/bin, or any permit-required confined space, none of the "just pop the cover and reseal it" advice applies without the full permit process. Those spaces need atmospheric testing before entry and carry engulfment risk — never enter a bin or silo under a live draw. **H₂S (hydrogen sulfide) kills fast and deadens your**

sense of smell around 100 ppm (IDLH ~100 ppm), so a "the rotten-egg smell went away" is a warning, not an all-clear. Methane is explosive from roughly 5% (LEL, Lower Explosive Limit) to 15% (UEL, Upper Explosive Limit) in air. Wear a bump-tested personal 4-gas monitor (H₂S, LEL, O₂, CO). Biogas areas are typically classified Class I Division 1/2 (or Zone 0/1/2) under NEC 500/505, which governs what enclosures and connectors are even legal there — you don't open a rated enclosure hot in a classified area to chase a CRC error. Get the permit, or get the space cleared and locked out first.

Check-order: physical layer first

Most comms faults are physical. Start here because it's the highest-probability, lowest-cost place to look, and because the CRC counters you already read point you straight at the bad run.

Patch and field cable. Follow the specific run that faulted, end to end. Look at the whole path: the patch cable at the switch, the field cable through the raceway, the connector at the device. In a plant the field cable takes the abuse — crushed in a cable tray, pinched by a panel door, baked against a hot pipe, or flexed on a cable carrier until a conductor work-hardens and cracks.

Connectors: RJ45 and M12. The RJ45 in a control panel is a liability — the latch breaks, the crimp backs out, dust and oil film get on the contacts. Field devices usually use M12 (the threaded circular connector, D-coded for 100 Mbit Ethernet, X-coded for gigabit) precisely because RJ45 doesn't survive vibration and washdown. Check that M12 rings are actually torqued, not finger-snug. A loose M12 is the single most common intermittent I find on packaging and process skids.

Bend radius and run length. Two limits people blow past: - Bend radius. Kinking a data cable tighter than roughly 4x its outside diameter (check the cable's own spec) changes the pair geometry and wrecks the impedance. A zip-tied 90-degree bend at a panel gland is a classic. - Run length. Copper Ethernet (Cat5e/Cat6) is a **100 meter (about 328 ft) channel limit, total**, patch cords included. I've seen 140-meter runs that "worked" until summer heat pushed the margin over the edge. If you're past 100 m, or crossing between buildings, that run wants fiber, not another coupler.

Copper vs. fiber. Fiber isn't just for distance. Any run that shares a raceway with power, crosses a ground-potential difference between buildings, or lives next to big drives is a candidate for fiber precisely because glass is immune to the electrical noise problems in the next section. If a segment only faults when the plant is running hard, and it's copper near power, fiber is often the real fix rather than another round of cable swaps.

Corrosion and moisture in a harsh plant. Washdown, condensation, and corrosive atmospheres (chlorine rooms, digesters, food acid) get into connectors. Green or white powder on RJ45/M12 contacts, or water beaded inside a connector boot, raises contact resistance and does it *intermittently* as temperature and humidity change. A connector that ohms out fine on a dry morning can fault every afternoon. Check IP-rating integrity: a cracked boot or a missing gland seal is the entry point.

Electrical noise and grounding

This is the layer that gets blamed last and causes the most maddening intermittents, because the symptom (CRC errors, connection timeouts) looks identical to a bad cable but the bad actor is three feet away.

VFD-induced noise. A VFD switches its output at high frequency to synthesize the motor waveform. That switching radiates and couples onto anything running parallel and close — including your data cable. The tell: **error counters that climb only when a specific drive runs, and specifically when it's loaded or ramping.** Idle plant, clean counters; start the big drive, CRC errors march. That's coupling, not a broken wire.

Fixes, in order: - **Separation.** Get data cable out of the same tray/conduit as drive output (motor) cables. A common rule of thumb is keep signal and power runs 8-12 inches apart for parallel runs and cross at 90 degrees where they must meet — verify against your site's cable-separation standard, but the principle is distance and perpendicular crossings. - **Shield ground — match the scheme to the protocol.** Shielded data cable only helps if the shield is grounded the way that cable and protocol actually call for, and the right answer isn't the same for every bus. High-speed industrial Ethernet (EtherNet/IP, PROFINET) and PROFIBUS are normally bonded **at both ends with a 360-degree connection to chassis** — at Ethernet frequencies the shield needs a low-impedance path to ground at each end to do its job, and a shielded connector that only clamps the drain wire isn't the same as a full circumferential bond. Some serial buses go the other way: DeviceNet, for one, grounds the drain at a **single point** near the power supply. The real trap is a ground-potential difference between two bonded ends driving current down the shield — a genuine problem on long runs between buildings, and the reason single-point (or a hybrid ground: solid at one end, small capacitor at the other) exists at all. Confirm what your specific cable and protocol call for and that the install matches it; mismatches here are extremely common after field additions. - Drive-side hygiene: proper motor cable (shielded VFD cable with the shield bonded 360 degrees at both drive and motor), line reactors or filters where specified, and a solid drive ground. A drive that's noisy on the network is often a drive that's grounded poorly at the motor.

Do this one change at a time and re-read the counters each time. Separation, then shield, then filtering — if you change three things at once you'll never know which one mattered, and you won't be able to hand the next tech a record.

Switch and topology

Managed vs. unmanaged. An unmanaged switch is a black box — it can't tell you anything, so on a plant network it's a place faults go to hide. This is the argument for managed switches: they give you the port counters and timestamps this whole method depends on. If a critical segment runs through an unmanaged switch and you're chasing a ghost, temporarily dropping in a managed switch (or a mirror port) to get visibility is often the fastest path to the answer.

Auto-negotiation, duplex, and speed mismatch. One of the most common and most misdiagnosed faults. If one end is set to auto-negotiate and the other is forced (hard-set to, say, 100/full), auto-neg fails to detect duplex and defaults to *half*. Now you have a **duplex mismatch**: full-duplex on one end, half on the other. It *works* at low traffic and falls apart under load with late collisions and CRC/alignment errors — looks exactly like a bad cable. Rule: either both ends auto, or both ends hard-set identically. Never mix. The switch port stats will show the collisions and the negotiated speed/duplex — read them.

Ring/redundancy recovery. Many plant networks run a redundant ring (DLR — Device Level Ring — on EtherNet/IP, MRP on PROFINET, or a proprietary ring). A ring is supposed to heal in milliseconds when one link breaks. Two failure modes: (1) the ring is broken in *two* places, so it can't heal, and (2) the ring healed silently around a real fault and nobody noticed, so you're one more break from an outage with no warning. Check the ring/redundancy status object — it tells you if a link is down and where. A "working" plant on a broken ring is a time bomb; the diagnostic that saves you is reading ring status *before* the second failure.

Broadcast/multicast storms and IGMP. Here's the EtherNet/IP-specific one that catches people. EtherNet/IP delivers its I/O (implicit) data as **multicast** by default — one producer, many consumers, sent to a multicast group. Without management, a switch floods multicast to every port like a broadcast. On a busy I/O network that flood can saturate links and CPUs and cause connection timeouts network-wide. The fix is **IGMP** (Internet Group Management Protocol — the mechanism switches use to learn which ports actually want a given multicast group) **snooping**, enabled on the managed switches with a querier defined. With IGMP snooping on, multicast goes only to ports that joined the group; without it, you get storms that look like random dropouts across unrelated devices. If a whole EtherNet/IP segment drops connections intermittently and no single cable is the culprit, check IGMP snooping and the querier first. (Newer devices can use unicast EtherNet/IP connections, which sidesteps this — but mixed networks still bite.)

A true broadcast storm — often a physical loop created by someone patching two switch ports together, or a failed NIC blasting frames — will show as sky-high broadcast counts and near-100% port utilization on the switch. The port stats find it in seconds.

Protocol and configuration

Once physical, noise, and switch layers are clean, the fault is in how the protocol is configured. These are the ones that don't show up until the network is otherwise healthy.

EtherNet/IP: RPI and connection timeouts. The **RPI** (Requested Packet Interval — how often, in milliseconds, a device is asked to produce its I/O data) sets the heartbeat of each connection. EtherNet/IP declares a connection timeout after a multiple of the RPI passes with no packet (commonly 4x RPI). Set the RPI too aggressively for the network and CPU load and you get connection timeouts under load — the network can't keep the heartbeat and the controller declares

the device gone even though nothing is broken. Symptom: connections drop when traffic peaks, recover when it settles. Check the configured RPI against the device count and switch load; loosening a too-tight RPI (say from 5 ms toward 10-20 ms, within what the process tolerates) often clears load-related timeouts. Read the connection timeout counters to confirm it's timeouts, not physical errors.

Modbus TCP. Simpler, and its faults are usually address or timing. Check: correct unit/slave ID, register addressing (the classic off-by-one from 0-based vs 1-based, and holding vs input register confusion), and the master's response timeout versus how long a loaded slave actually takes to answer. A slave that occasionally answers slowly under load will throw intermittent timeouts that look like comms drops. Also watch for two masters polling one slave and stepping on each other.

PROFINET and PROFIBUS: termination and addressing. - PROFINET rides Ethernet, so everything above applies, plus device naming: each device has a **device name** the controller uses to find it. A duplicated or missing device name, or a wrong DCP assignment after a device swap, drops that device. The controller diagnostic buffer names the offending station. - PROFIBUS is a different animal — an RS-485 multidrop bus. It **must be terminated at both physical ends** and only the ends. A missing or extra termination, or termination powered off, garbles the whole segment. Each station needs a unique address, no duplicates, and every station must be set to the same baud. A single wrong address or a bad terminator can take down every device downstream — the tell is that the fault isn't isolated to one node.

DeviceNet: baud and termination. A CAN-based (Controller Area Network) differential bus — not RS-485, though it's the same multidrop discipline. **Every node on the segment must be at the same baud rate** — one node set wrong can disrupt the bus — and the trunk needs a 120-ohm terminating resistor at *each* end, exactly two. Check trunk length limits for the chosen baud (lower baud allows longer trunk; verify against the DeviceNet spec table for your rate). Duplicate MAC IDs (node addresses) also cause conflicts. On these buses, "termination and addressing" is 90% of the config faults.

Isolate by segment

When the logs point at a region but not a device, bisect. Split the suspect segment: disconnect half the devices (or open a managed switch's ports in groups) and see if the fault clears. Halve again. On a ring, open it deliberately at a known point to force traffic one way and localize. Each split, re-read counters. You're converging on the one run, connector, or node that's dirty — and doing it with data instead of a box of spare parts.

Always correlate with load. Bench-testing a suspect run in a quiet plant proves nothing if the fault only appears with the VFD running and the line at full rate. **Watch the error counters climb under load, or with the drive running** — recreate the condition, don't just poke at it cold.

Intermittent faults: temperature, vibration, water

The dropouts that survive a day of troubleshooting are almost always environmental, because they only appear under the condition you're not testing in.

- **Temperature.** Connectors and marginal cable expand and contract. A run that's fine at 6am morning startup faults every afternoon when the building heats or a nearby process runs hot. Correlate fault timestamps against time-of-day and against a heat source cycling.
- **Vibration.** Anything on a motor, pump, mixer, conveyor, or press vibrates connectors loose over weeks. An M12 that was finger-tight walks out; an RJ45 latch fatigues and backs off. Vibration faults track with machine run-time and get worse over days.
- **Water ingress.** Washdown and condensation get into connectors and junctions and change contact resistance with humidity. Faults cluster after washdown or on humid days, then clear when things dry. Look for water in boots and glands, and for the green/white corrosion it leaves behind.

The common thread: an intermittent that correlates with a *condition* (heat, run-time, washdown) is environmental, and the fix is at the physical connector or the cable route, not in the config.

Symptom -> likely cause -> check (and log)

Symptom	Likely cause	Check (and log)
CRC/alignment errors climbing on one switch port	Bad cable, loose/corroded connector, or coupled noise on that run	Read port CRC count now and again in 10 min; follow that one run end to end; reseal/ring out connectors. Log counter rate + which run.
Errors climb only when a specific VFD runs	VFD noise coupling onto parallel data cable; shield/ground wrong	Read counters drive-off vs drive-running; verify separation and single-point shield ground. Log delta and drive ID.
Connections drop under peak traffic, recover when quiet	RPI too tight for load, or multicast storm (no IGMP snooping)	Read connection-timeout counters; check RPI vs device count; verify IGMP snooping + querier on switches. Log RPI and timeout count.
Random drops across unrelated Ethernet/IP devices	Multicast flooding / broadcast storm	Check switch broadcast/multicast counts and port utilization; confirm IGMP snooping; hunt a physical loop. Log utilization %.
Link "works" but collisions + CRC under load	Duplex/speed mismatch (one auto, one forced)	Read negotiated speed/duplex on both ends; set both auto or both hard-set the same. Log both port settings before/after.

Symptom	Likely cause	Check (and log)
Whole PROFIBUS/DeviceNet segment down, not one node	Missing/extra termination, duplicate address, or baud mismatch	Verify exactly two terminators at physical ends; check each node's address is unique and baud matches. Log addresses/baud/term state.
One PROFINET device won't connect after a swap	Wrong/duplicate/missing device name	Read CPU diagnostic buffer for the station name; reassign device name via DCP. Log old/new name.
Fault appears afternoons / hot days only	Temperature on a marginal connector/cable	Correlate fault timestamps vs time-of-day and heat source; inspect the run for heat exposure. Log timestamps.
Fault worsens over days on a vibrating machine	Vibration loosening a connector	Check M12 torque / RJ45 latch on that machine; correlate with run-time. Log which connector, torque state.
Faults cluster after washdown / humid days	Water ingress / connector corrosion	Inspect boots and glands for water; check contacts for corrosion; verify IP seal. Log where water found.
Ring "healthy" but a link is actually down	Ring already healed around one break	Read ring/redundancy status object; find and fix the down link before a second break. Log the down link location.

10-minute triage

The first ten minutes, from an engineering laptop and an operator terminal, before you open anything:

- Minute 0-2: Read the controller.** Pull the module fault / connection status bits and the fault log or diagnostic buffer. Write down *which* connection faulted and the *timestamp*.
- Minute 2-4: Read the switch.** On the managed switch in that segment, pull per-port stats. Note CRC/error counts, link flap counts, last-flap timestamp, and negotiated speed/duplex on the suspect port.
- Minute 4-5: Line up the times.** Does the controller fault timestamp match a switch port flap? Match a drive event? A matching triple locates your segment.
- Minute 5-7: Is it load-related?** Note plant state at the fault times. Faulting under load / with a drive running points to noise, RPI, or a storm. Faulting cold and random points to physical/connector.
- Minute 7-9: Take a rate, not a snapshot.** Re-read the key counters. Are CRC errors *still climbing* right now? A climbing counter means the fault is live and you can chase it in real time. A frozen

counter means it's intermittent — go to the condition (heat, washdown, vibration, specific drive) that triggers it.

6. **Minute 9-10: Decide the next hands-on step and the safety it needs.** If the trail says "reseat the M12 on pump 3" or "check the shield ground at the drive," that's now a LOTO / energized-work task — stop, apply the right zero-energy state and PPE, and do it as a recorded single change. If it says "IGMP snooping is off" or "RPI is 4 ms," that's a config change you can stage and schedule, one change at a time, with a record.

Everything in the first nine minutes is read-only. You should know your segment and probably your cause before a single connector is touched.

A closing note, plainly: comms dropouts are where an independent second look earns its keep, because these faults usually live in the seam between vendors — the drive integrator's cable tray crossing the network integrator's data run, with the field devices added by a third crew. Each vendor's gear checks out on its own bench. The fault is in the handoff, and the record — counters, timestamps, one change at a time — is what finally pins it.